A FINITE STABLY FREE MODULE MODULO AN IDEAL IN THE JACOBSON RADICAL LIFTS TO A FINITE STABLY FREE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tags are carried by claim_key and by the external references. The same-chapter pointer lemma-isomorphic-finite-projective-lifts (0BC6) is counted by a sibling card of this batch and is not invoked anywhere in the counted spans.

Allowed standard prerequisites (editorial).

- Determinant of an endomorphism of a finite free module whose reduction modulo the ideal is the identity. used at proof lines 107-109; the same fact is packaged in the hypothesis that every element of 1 + I is a unit. The determinant of the lifted composite is congruent to 1 modulo I and is therefore invertible, so the lifted composite is itself invertible; that invertibility is what makes the sequence 0 -> Ker(phi) -> R^m -> R^n -> 0 split.
- Finitely generated projective modules over a ring and the splitting of short exact sequences of projective modules. the chapter's own remark at source lines 66-67 that a stably free module is projective. Makes the kernel of the split surjection phi a direct summand of the finite free module R^m, hence finite stably free.
- Stable freeness and stable isomorphism of modules over a ring. the chapter's own Definition definition-stably-free at source lines 53-64, which fixes the meaning of the terms used in the statement. Fixes the class of modules being lifted: E is finite stably free over R/I and the constructed M must be finite stably free over R with M/IM isomorphic to E.

Source reference. Original lemma 00NZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 877–877. Cross-chapter pointer into the Algebra chapter; it is the standard source of a reduction and the argument built on top of that reduction stays inside the counted proof span.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Cross-chapter pointer into the Algebra

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chapter supplying a standard criterion of finite projectivity; it is background terminology, not part of the counted argument.

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Cross-chapter pointer into the Algebra chapter for Nakayama's lemma, a standard prerequisite; both applications of it and the surrounding argument stay inside the counted proof span.

Source reference. Original lemma 05G8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7052–7052. Cross-chapter pointer into the Algebra chapter for the fact that a surjective endomorphism of a finite module is an isomorphism, used only to pass from surjectivity to invertibility.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Assume that every element of $1 + I$ is a unit (in other words I is contained in the Jacobson radical of R). For every finite stably free R/I-module E there exists a finite stably free R-module M such that $M/IM \cong E$.*

Proof. Choose a n and m and an isomorphism $E \oplus (R/I)^{\oplus n} \cong (R/I)^{\oplus m}$. Choose R-linear maps $\varphi : R^{\oplus m} \to R^{\oplus n}$ and $\psi : R^{\oplus n} \to R^{\oplus m}$ lifting the projection $(R/I)^{\oplus m} \to (R/I)^{\oplus n}$ and injection $(R/I)^{\oplus n} \to (R/I)^{\oplus m}$. Then $\varphi \circ \psi : R^{\oplus n} \to R^{\oplus n}$ reduces to the identity modulo I. Thus the determinant of this map is invertible by our assumption on I. Hence $P = \mathrm{Ker}(\varphi)$ is stably free and lifts E. $\square$